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Healthy lungs contain large amounts of elastic connective tissue around the alveoli, containing elastin, which makes the lung tissue elastic. People with emphysema and bronchitis have difficulty in exhaling because the enzyme elastase destroys the elastin around the alveoli and reduces the elasticity of the lungs.

- **Inspiratory Reserve volume (IRV)** Additional volume of air a person can inspire by forceful inspiration is called Inspiratory Reserve Volume. The normal value is 2500–3000 mL.
- **Expiratory Reserve volume (ERV)** Additional volume of air a person can forcefully exhale by forceful expiration is called Expiratory Reserve Volume. The normal value is 1000–1100 mL.
- **Residual Volume (RV)** The volume of air remaining in the lungs after a forceful expiration. It is approximately 1100–1200 mL.

Respiratory capacities:

- **Vital capacity (VC)** the maximum volume of air that can be moved out during a single breath following a maximal inspiration. A person first inspires maximally then expires maximally. VC=ERV+TV+IRV
- **Inspiratory capacity (IC)** The total volume of air a person can inhale after normal expiration. It includes tidal volume and inspiratory reserve volume. IC=TV+IRV
- **Expiratory capacity (EC)** The total volume of air a person can exhale after normal inspiration. It includes tidal volume and expiratory reserve volume. EC=TV+ERV
- **Total Lung Capacity (TLC)** The total volume of air which the lungs can accommodate after forced inspiration is called Total Lung Capacity. This includes the vital capacity and the residual volume. It is approximately 6000mL. TLC=VC+RV
- **Minute Respiratory Volume** The amount of air that moves into the respiratory passage per minute is called minute respiratory volume.

  Normal TV = 500mL; Normal respiratory rate = 12 times/minute
  Therefore, minute respiratory volume = 6 Litres/minute (for a normal healthy man).

Dead space

Some of the inspired air never reaches the gas exchange areas but fills the respiratory passages where exchange of gases does not occur. This air is called dead space.

Dead space is not involved in gaseous exchange. It amounts to approximately 150mL.

6.4 Exchange of gases

The primary site for the exchange of gases is the alveoli. The uptake of O_2 and the release of CO_2 occur between the blood and tissues by simple diffusion driven by partial pressure gradient of O_2 and CO_2. Partial pressure is the pressure contributed by an individual gas in a mixture of gases. It is represented as pO_2 for oxygen and pCO_2 for carbon–